TECHNICAL ROADMAP TEMPLATE

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Naming Convention: PH_CIT-U_Odd-1-Out_Roadmap.pdf

TEAM INFORMATION

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Track	Telemedicine
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SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

In 2021, 2,478 Filipino mothers died from pregnancy-related causes, a tragedy disproportionately affecting Geographically Isolated and Disadvantaged Areas (GIDA), where emergency transport costs up to ₱6,000. While the Department of Health has deployed billions in portable ultrasound devices, a severe specialist shortage leaves many of these assets idle; as of late 2025, only 200 of 600 HFEP-funded health centers were fully functional. Kalinga AI solves this "Infrastructure Paradox." We deploy an offline-first, AI-assisted diagnostic triage system that empowers rural midwives to capture specialist-grade ultrasound data. Operating via an asynchronous "store-and-forward" pipeline, on-device AI guides the hardware sweep, selects critical frames, and generates preliminary risk scores (e.g., preeclampsia probability). Highly compressed data is then synced to a cloud queue for remote verification by OB-Gynes. By activating dormant government equipment, Kalinga AI prevents maternal deaths, eliminates systemic e-waste, and creates a scalable, asset-light triage model replicable across ASEAN.

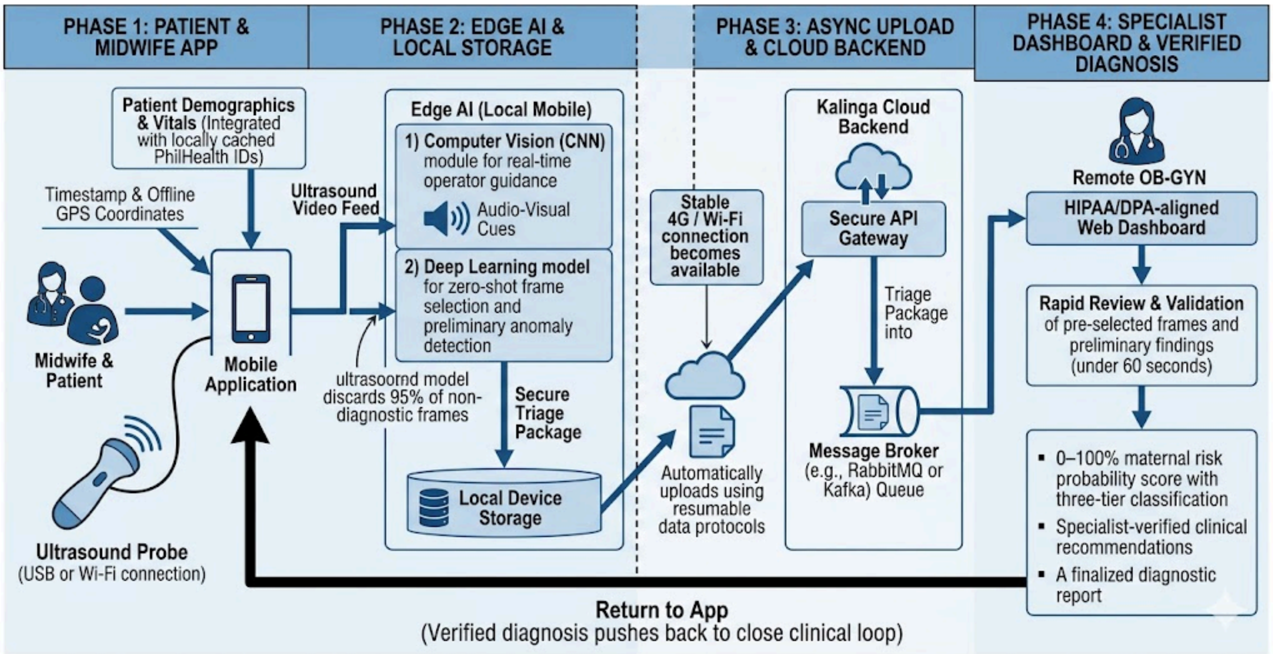
SECTION 2: TECHNICAL ARCHITECTURE

2.1 System Components

- **Inputs**
    - Ultrasound video feed generated by a handheld probe connected via USB or Wi-Fi.
    - Patient demographics and vitals (integrated with locally cached PhilHealth IDs).
    - Timestamp and offline GPS coordinates for epidemiological mapping.
- **Processing Core**
    - **Edge AI (Local Mobile):** Runs a Computer Vision (CNN) module for real-time operator guidance, and a Deep Learning model for zero-shot frame selection and preliminary anomaly detection, entirely offline.
    - **Cloud Backend:** Secure API gateways handle asynchronous uploads. A message broker (e.g., RabbitMQ or Kafka) manages the queue of incoming triage packages.
    - **Human-in-the-Loop:** A HIPAA/DPA-aligned Web Dashboard enables remote OB-GYNs to rapidly review and validate the AI's preliminary findings.

- Outputs
  - A 0–100% maternal risk probability score with three-tier classification.
  - Specialist-verified clinical recommendations routed back to the Barangay Health Center (BHC).
  - A finalized diagnostic report routed directly back to the midwife's mobile application.

2.2 Architecture Diagram:



SECTION 3: AI APPROACH & MODEL SELECTION

AI Approach & Model Selection

- **Primary AI Types**
  - **Computer Vision (CNN):** Custom lightweight CNNs based on MobileNetV3, deployed via PyTorch Mobile / ONNX runtime. This ensures the model is small enough to run inference on mid-tier Android devices without draining the battery or causing thermal throttling.
  - **Deep Learning:** Extracts features and performs zero-shot plane classification for biometry.
  - **Predictive Models:** Identifies high-risk markers for conditions like preeclampsia.
- **Model Selection**
  - **TensorFlow Lite / PyTorch Mobile:** Deployed strictly at the edge for offline inference.
  - **FetalCLIP Framework:** Serves as the foundation for ultrasound analysis and frame selection.
  - **Cloud Validation Pipelines:** Secure environments separate from deployment use verified data to retrain models.
- **Reasoning**
  - **Edge AI Criticality:** : We practice "automation without replacement." Edge AI is critical because GIDA clinics lack the bandwidth for live video telemedicine (like Starlink). On-device processing is mandatory for immediate guidance.
  - **Asynchronous Supremacy:** Synchronous telemedicine fails at scale due to the severe OB-GYN shortage. Store-and-forward optimizes specialist time.

  - **Human Validation:** AI synthesizes dozens of variables in seconds, but to resolve medical liability and ensure LGU trust, the human-in-the-loop framework ensures an OB-Gyne makes the final clinical decision.

SECTION 4: DATA STRATEGY & ETHICS

- **Data Sources:** System training utilizes open-source multimodal datasets including FetalCLIP and NatalIA PBF-US1. Clinical logic follows established DOH guidelines and NPC Advisory frameworks.
- **Data Cleaning:** We strictly utilize datasets annotated by medical professionals. Tabular data for the risk prediction model will undergo rigorous outlier detection to ensure synthetic or anomalous vital signs do not skew the baseline.
- **Licensing:** Deployment utilizes compliant open-source research models. Handling aligns with the Data Privacy Act of 2012.
- **Ethics & Informed Consent:** Consent is logged digitally before scanning. We ensure equitable access by providing plain-language verbal explanations in regional dialects, supported by printed summaries using visual icons for low-literacy patients. All transmitted data is encrypted and de-identified within the clinical chain.
- **Bias Mitigation:**
  - *Strategy 1 (Geographic Diversity):* We actively source training data from rural settings to counter urban-centric dataset bias.
  - *Strategy 2 (Hardware Agnosticism):* Models are trained against a variety of transducer frequencies to ensure performance across different idle DOH hardware tiers.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

Phase	Activity / Task	Tools Used	Expected Outcome
Sprint 1 (May 1-15)	**Project Lead / Researcher:** Design & distribute an inventory audit form to target rural BHCs via local networks to catalog active/idle ultrasound probe specifications (frequencies, connection types). **AI Engineer:** Clone FetalCLIP repository, establish local environment dependencies, and parse open-source datasets (NatalIA PBF-US1) for preprocessing. **AI Engineer & Developer:** Conduct data cleaning: Apply cropping, pixel normalization, and contrast-limited adaptive histogram equalization (CLAHE) to filter artifact-heavy scan frames.	Google Forms, Excel GitHub, Miniconda, PyTorch Python (OpenCV, NumPy) Figma	Hardware compatibility matrix mapping at least 15 local GIDA health center assets. Working local repository with clean data loaders executing without syntax errors. A curated, structured training dataset of ultrasound images. Interactive low-fidelity prototypes detailing user authentication and scan submission flows.

	UI/UX Designer: Wireframe the baseline user journey for both the midwife's mobile application and the remote OB-Gyne dashboard based on DOH workflow standards.		
Sprint 2 (May 16-31)	**AI Engineer:** Convert and optimize the FetalCLIP model and the baseline CNN probe-guidance network into a quantized flatbuffer format suitable for mobile chips. **Mobile Developer:** Develop the Android camera preview interface to capture live frames from digital USB/Wi-Fi ultrasound probe SDK feeds. **AI Engineer & Developer:** Bind the .tflite models into the app backend to trigger real-time, on-device audio/visual guidance cues (e.g., directional arrows for probe adjustments). **Project Lead (Domain):** Code the localized mathematical risk scoring algorithm using raw parameters (maternal age, BMI, systolic/diastolic blood pressure history).	TensorFlow Lite, ONNX Android Studio, Kotlin Android Studio, TFLite Interpreter Java/Kotlin	Compiled .tflite model files compressed to under 45MB with less loss in plane classification accuracy Native Android application layout capable of rendering real-time mock ultrasound video streams at 30 FPS. UI layout that dynamically overlays text/audio instructions based on edge classification confidence scores. Deterministic risk calculator module outputting low/medium/high preeclampsia alerts via local data matrix lookup.
Sprint 3 (June 1-15)	**Mobile Developer:** Build the localized encrypted storage schema to securely queue patient records, image packages, and telemetry data on the mobile device during offline states. **Full-Stack Developer:** Implement the auto-resumable background sync service that triggers data packets upload via secure connection layers whenever a network signal is discovered. Architect the web-based backend server queue to handle incoming asynchronous triage	SQLite, Room DB, SQLCipher Node.js, Axios, HTTPS/TLS Node.js, Express, RabbitMQ React, Tailwind CSS	Encrypted local database holding up to 100 complete, un-synced patient scan packages offline. Functional background service script that transmits compressed ZIP data packages to the backend queue. Scalable server API endpoints logging and organizing data payloads

	packages without dropping payload data. **UI/UX Designer & Dev:** Code the front-end interface for the OB-Gyne review screen, enabling specialists to display frames, check patient history, and toggle verification results.		in a centralized relational index. High-fidelity specialist portal rendering diagnostic frame queues with a 60-second processing checkout layout.
Sprint 4 (June 16-25)	**Whole Team:** Execute local user simulation testing: Intentionally disconnect mobile devices, input dummy client data, re-establish signal, and verify backend queue persistence. **Designer:** Conduct a limited walkthrough simulation of the user interface with local nursing/midwifery students to validate screen readability and dialect ease. **AI Engineer & Developer:** Generate the official release build of the mobile system and deploy cloud servers for virtual evaluation availability. **Project Lead:** Script, direct, and record a highly engaging 3-to-5 minute video demonstrating a live end-to-end "blind sweep" to backend verification pitch.	Android Emulator, Postman Google Meet / Figma Gradle, Render/AWS, TestFlight Loom, Premiere/CapCut	Comprehensive edge-to-cloud verification log proving zero data corruption across 50 simulated test packages. Usability feedback scorecard and subsequent rapid minor revisions to the app's visual icons and local phrasing. Production-ready Android .apk package file downloadable for hackathon review panels. Polished, high-definition final demo video file uploaded to the official P2A portal ahead of the June 25 deadline.

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

- **Scaling Potential:** Kalinga AI ensures financial sustainability through integration with PhilHealth Konsulta reimbursement packages, creating a self-sustaining revenue loop. Community uptake is driven by the "Magnet Effect," using "Heartbeat Demo Days" to boost antenatal care compliance. Because the software is hardware-agnostic, it can rapidly expand across similar archipelagic and highland healthcare systems in Indonesia and Vietnam without requiring massive new hardware procurement.
- **Technical Constraints:**
  - *Model Accuracy Limitations:* Variations in maternal BMI can affect ultrasound wave penetration, requiring edge model confidence thresholds to be carefully calibrated.
  - *Absolute Dead Zones:*The store-and-forward architecture handles intermittent signals seamlessly, but severe dead zones still require the physical transport of the mobile device to a connected hub to sync triage packages.
  - *Hardware Variability:* The system relies on the distribution density of modern digital/handheld probes; older, immovable CRT-based ultrasound units lacking digital outputs cannot be integrated.